

Development Application Profile

Nova Fleet Insights Platform

DEMONSTRATION DOCUMENT All names, identifiers, contacts, systems, metrics and dates in this document are fictional and intended only for application KT planning demonstrations.

Document control	Value
Document ID	DEMO-DEV-NFIP-002
Version	1.0
Document owner	Aditi Menon (fictional)
Classification	Internal - Demo
Prepared for	Application Knowledge Transfer Planning
Status	Draft for demonstration

Application Transition & Knowledge Transfer Profile

1. Executive Summary

Nova Fleet Insights Platform is a fictional cloud-native product that provides fleet utilization, energy-consumption and maintenance insights. The development transition covers product architecture, backlog delivery, engineering standards, test automation and release governance.

Transition objective Enable the receiving engineering squad to independently design, implement, test and release product increments with traceable quality and security controls.

2. Application Setup

Field	Value
Application Name	Nova Fleet Insights Platform
Service Model	Product Development - Build and Release
Business Domain	Connected Fleet Analytics
Business Owner	Lukas Steiner (fictional)
Application Owner	Aditi Menon (fictional)
Service Criticality	Strategic Product - Pre-production
User Footprint	Pilot cohort of 320 users
Primary Regions	Germany, India and Spain
KT Start Date	12-Oct-2026
Timezone	IST with CET overlap
Working Days	Monday-Friday
Working Hours	09:00-18:00 IST
Follow-up Sessions	4
Shadow Days	5
Reverse Shadow Days	5

3. Scope and Service Boundaries

In Scope

- Product backlog refinement and feature delivery
- Frontend, API and analytics service development
- Automated testing, code quality and dependency scanning
- CI/CD pipelines and deployment to development, test and staging
- Architecture decisions, coding standards and release documentation

Out of Scope

- 24x7 production operations and end-user service desk
- Commercial product pricing and sales enablement
- Source telemetry hardware firmware
- Corporate tenant and network administration

4. Team and Governance

Role	Name	Location	Responsibility
Product Owner	Aditi Menon	Bengaluru	Product outcomes, priorities and acceptance
Domain SME	Lukas Steiner	Munich	Fleet analytics and business rules
Tech Lead	Diego Morales	Madrid	Architecture and engineering quality
QA Lead	Priya Deshpande	Pune	Test strategy and quality gates
DevOps Engineer	Omar Sheikh	Gurugram	Pipelines, environments and observability

Governance Cadence

Forum	Cadence	Participants	Purpose
Daily Stand-up	Weekdays	Delivery squad	Progress, blockers and flow
Refinement	Weekly	PO, SMEs, squad	Ready backlog with acceptance criteria
Architecture Forum	Biweekly	Tech leads	Decisions, risks and standards
Sprint Review	Biweekly	Stakeholders	Demo outcomes and collect feedback

5. Architecture and Technology Profile

Layer	Technology / Service	Operational Notes
Frontend	React 19 / TypeScript	Micro-frontend shell with Fluent UI components
API	Python 3.13 / FastAPI	Async APIs, OpenAPI contracts and Pydantic validation
Analytics	Databricks / Python	Curated utilization and energy models
Database	Azure Cosmos DB	Telemetry summaries partitioned by fleet ID
Cache	Azure Cache for Redis	Dashboard query and reference-data caching
Platform	AKS / Helm	Separate dev, test and staging namespaces
Delivery	Azure DevOps Pipelines	Build, scan, test, sign and deploy stages

Interfaces and Data Flows

Interface ID	System	Direction	Protocol	Frequency	Owner
IF-01	Telemetry Lake	Inbound	Event Hubs	Streaming	Data Platform
IF-02	Vehicle Master	Inbound	REST/OAuth2	Hourly	Master Data
IF-03	Identity Service	Inbound	OIDC	On login	IAM Team
IF-04	Reporting API	Outbound	REST/API key	On demand	Product Team

6. Service Targets and Controls

Measure	Target	Threshold / Control	Reporting
Build success	>= 95%	Main branch rolling 30 builds	Sprint dashboard
Unit coverage	>= 80%	New or changed code	Pull request gate
Critical defects	0 open	Before release candidate approval	Release checklist
API p95	< 600 ms	Staging load test at expected peak	Test report

Monitoring and Alerting

- Pipeline telemetry tracks lead time, failure rate and flaky tests
- Application Insights captures API dependencies and exceptions in test and staging
- Cost dashboard highlights namespace, Databricks and Cosmos DB consumption
- Synthetic dashboard journey runs every 15 minutes in staging

7. Knowledge Transfer Topic Plan

#	Topic	Category	Primary SME	Planned Outcome
1	Product vision and personas	Business	Aditi Menon	Prioritize outcomes and interpret acceptance
2	Fleet analytics rules	Functional	Lukas Steiner	Explain KPI calculations and edge cases
3	Frontend architecture	Technical	Diego Morales	Create and test a micro-frontend feature
4	FastAPI design standards	Technical	Diego Morales	Implement a versioned API with validation
5	Automated test strategy	Technical	Priya Deshpande	Run pyramid checks and quality gates
6	CI/CD and environments	Technical	Omar Sheikh	Promote a signed build to staging

8. Operational / Delivery Workflows

Feature Delivery

- 1. Refine the user story with measurable acceptance criteria and non-functional needs.
- 2. Record an architecture decision when the change affects shared contracts or security.
- 3. Implement behind a feature flag with unit, integration and contract tests.
- 4. Complete peer review, automated controls and product-owner acceptance.
- 5. Promote the immutable artifact and publish release notes.

Defect Triage

- 1. Reproduce and classify the defect using agreed severity criteria.
- 2. Link evidence, affected version and test case to the work item.
- 3. Prioritize the fix, validate regression coverage and close with release reference.

9. Risks, Dependencies and Readiness

ID	Type	Description	Impact	Mitigation / Next Step	Owner
R-01	Risk	Telemetry schema versioning is not enforced upstream	High	Add contract registry and compatibility tests	Tech Lead
R-02	Dependency	Pilot vehicle simulator has limited concurrency	Medium	Create deterministic replay datasets	QA Lead
R-03	Risk	Two analytics notebooks contain manual steps	Medium	Convert notebooks into parameterized jobs	Data SME

Readiness Checklist

Area	Status	Evidence / Comment
Backlog	Complete	Two sprints of refined stories in the demo scenario
Repositories	Complete	Branch policies and CODEOWNERS configured
Environments	Complete	Dev, test and staging access validated
Test automation	In progress	Performance baseline pending
Release rehearsal	Planned	Staging release and rollback exercise scheduled

10. Planned Holidays and Coverage Notes

Date	Coverage Note
02-Oct-2026	India support coverage through on-call roster; planned KT paused.
20-Oct-2026	Regional holiday coverage; P1/P2 support only.
25-Dec-2026	Global holiday; change freeze and emergency support model.

11. Acceptance Criteria

- Receiving squad completes one feature from refinement through staging release.
- All pull-request gates run successfully on the reference implementation.
- Architecture and API standards are applied without corrective comments.
- Rollback and feature-flag disablement are demonstrated in staging.

Approval note This demonstration profile is considered KT-ready when mandatory application setup fields are complete, topic owners are assigned, critical access is validated, and evidence links are recorded.